Chapter 16, Problem 78P

Problem

The transfer function of a certain circuit is

$$H(s) = \frac{10}{s+1} - \frac{6}{s+2} + \frac{12}{s+4}$$

Find the impulse response of the circuit.

Step-by-step solution

Step 1 of 3

Consider the transfer function of the circuit is,

$$H(s) = \frac{10}{s+1} - \frac{6}{s+2} + \frac{12}{s+4} \quad \text{...... (1)}$$

Definition of impulse response:

It is the output response of a circuit when the input is an impulse function $\delta(t)$.

The Laplace transform of an impulse function $\delta(t)$ is 1 $[\mathcal{L}[\delta(t)] = 1]$. So, the Impulse response is same as the inverse transform of the transfer function $H(s)$.

Step 2 of 3

Determine the impulse response of the circuit.

$$\begin{aligned} h(t) &= \mathcal{L}^{-1}[H(s)] \\ &= \mathcal{L}^{-1}\left[\frac{10}{s+1} - \frac{6}{s+2} + \frac{12}{s+4}\right] \\ &= 10\mathcal{L}^{-1}\left(\frac{1}{s+1}\right) - 6\mathcal{L}^{-1}\left(\frac{1}{s+2}\right) + 12\mathcal{L}^{-1}\left(\frac{1}{s+4}\right) \end{aligned}$$

Step 3 of 3

Use the following Inverse-Laplace transform formula.

$$\mathcal{L}^{-1}\left(\frac{1}{s+1}\right) = e^{-t}u(t)$$

Write the impulse response of the circuit.

$$\begin{aligned} h(t) &= 10e^{-t}u(t) - 6e^{-2t}u(t) + 12e^{-4t}u(t) \\ &= \left[10e^{-t} - 6e^{-2t} + 12e^{-4t}\right]u(t) \end{aligned}$$

Therefore, the impulse response of the circuit is $\boxed{\left[10e^{-t} - 6e^{-2t} + 12e^{-4t}\right]u(t)}$.